Day 01 Report: Titanic Survival Prediction

Oct 29, 2026 · @Hiru

Summary

XGBoost predicted Titanic survival best, with 84.4% accuracy in 5-fold cross-validation (752 of 891 passengers correct).

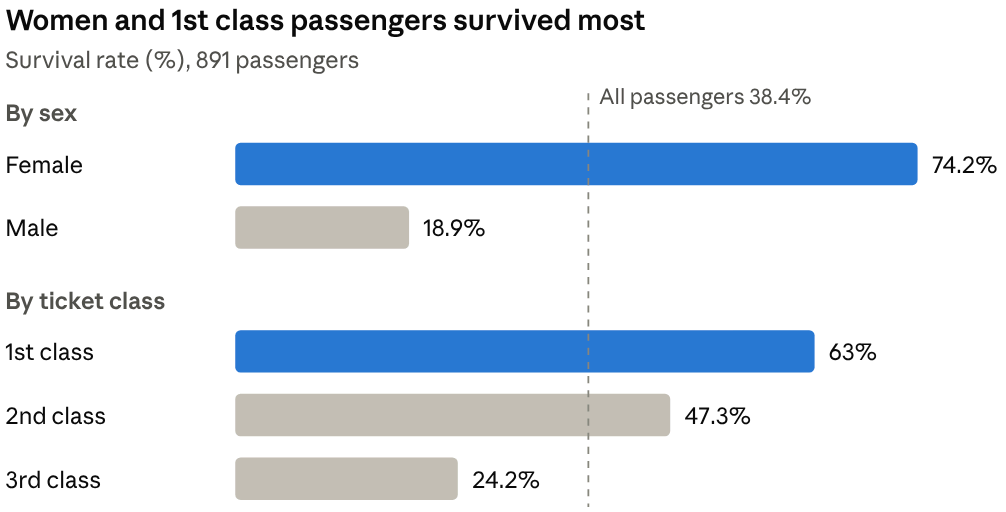
- Women, children and 1st class passengers were far more likely to survive.
- Adult men (title Mr) had the lowest survival, at 16%.
- The model's main weakness: it misses about 1 in 4 survivors.

Data

The Kaggle Titanic training set has 891 passengers and 12 columns; 342 survived (38%).

Column	Meaning	Missing
Survived	0 = died, 1 = survived (the target)	0
Pclass	Ticket class: 1, 2 or 3	0
Sex	male / female	0
Age	Age in years	177
SibSp / Parch	Siblings or spouses / parents or children on board	0
Fare	Ticket price	0
Cabin	Cabin number	687 (77%)
Embarked	Port: C, Q or S	2

What the data shows



Kaggle Titanic train.csv · 891 passengers

Sex and ticket class were the strongest signs of survival, reflecting "women and children first" and better access to lifeboats for richer passengers. Young boys (title Master) also survived at 58%, compared with 16% for adult men.

Cleaning and features

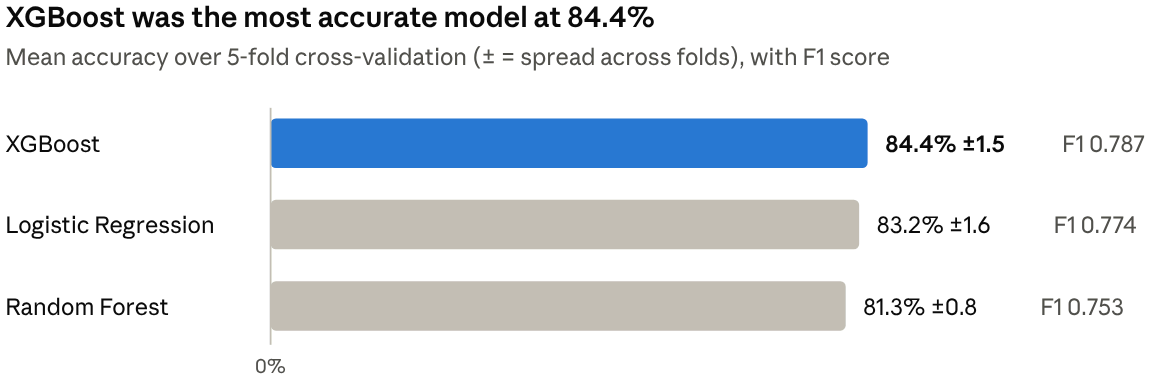
Missing values were filled with sensible guesses instead of dropping passengers.

Step	What was done	Why
Title	Took Mr, Mrs, Miss, Master from the name; rare titles grouped as Rare	Titles show age group and sex
Age	Filled with the median age for each title	A boy (Master) gets about 4, not the overall 28
Cabin	Turned into HasCabin (1 = yes, 0 = no)	77% missing; guessing adds noise, but having a cabin signals wealth
FamilySize	SibSp + Parch + 1	Alone vs. with family affected survival
Embarked	Filled with the most common port (inside the pipeline)	Only 2 missing

All scaling and encoding sits inside a scikit-learn Pipeline, so each test fold never leaks into training.

Models and results

Three models were tested with the same cleaning pipeline: Logistic Regression (one straight line), Random Forest (300 trees voting) and XGBoost (300 trees, each fixing the last one's mistakes). Each was trained on 4 of 5 groups and tested on the fifth, five times.

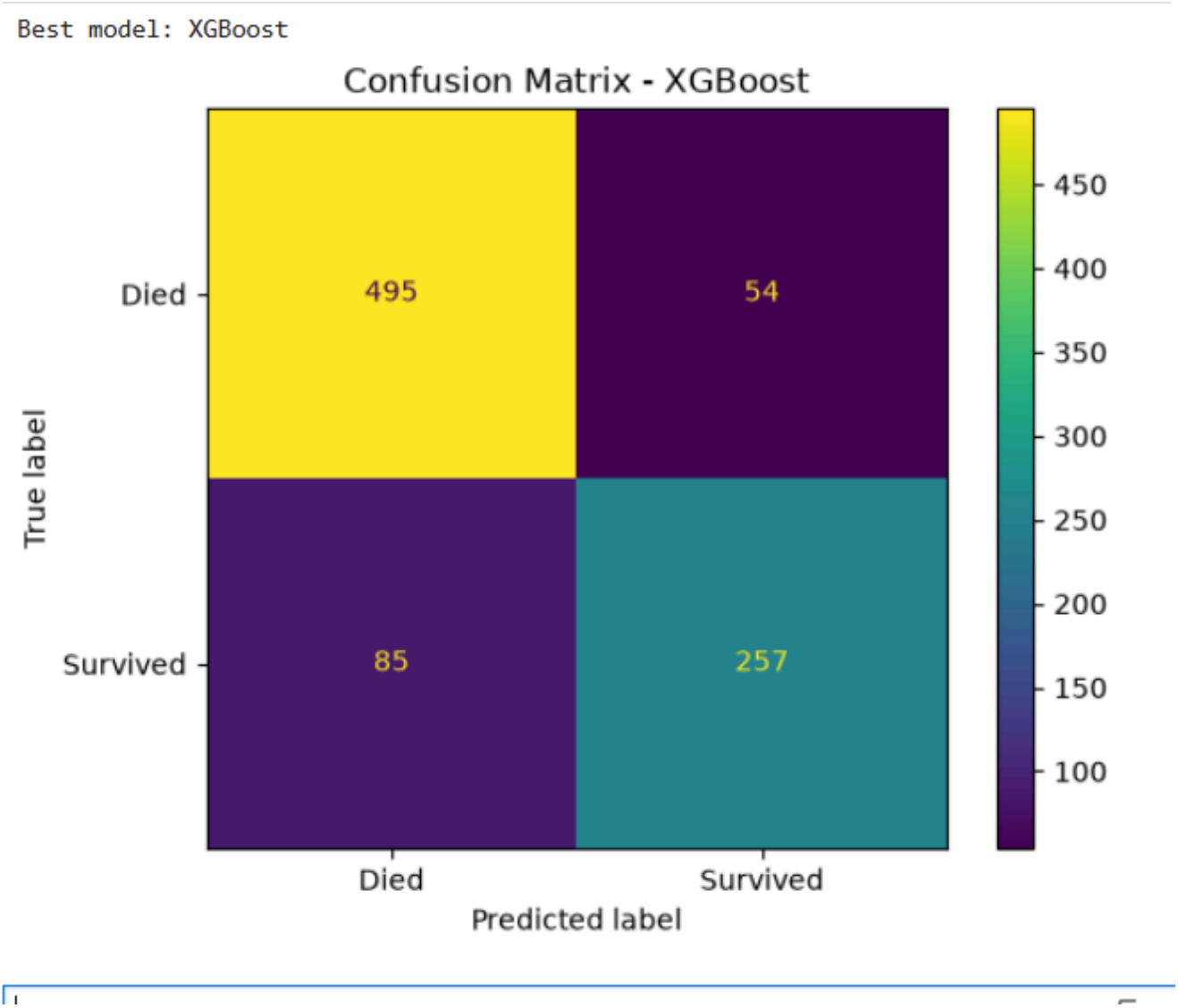


titanic.ipynb · 5-fold stratified cross-validation, random_state 42

All three are close, within about 3 points. XGBoost has the best accuracy and F1; Random Forest is the most stable across folds but the least accurate.

Best model in detail: XGBoost

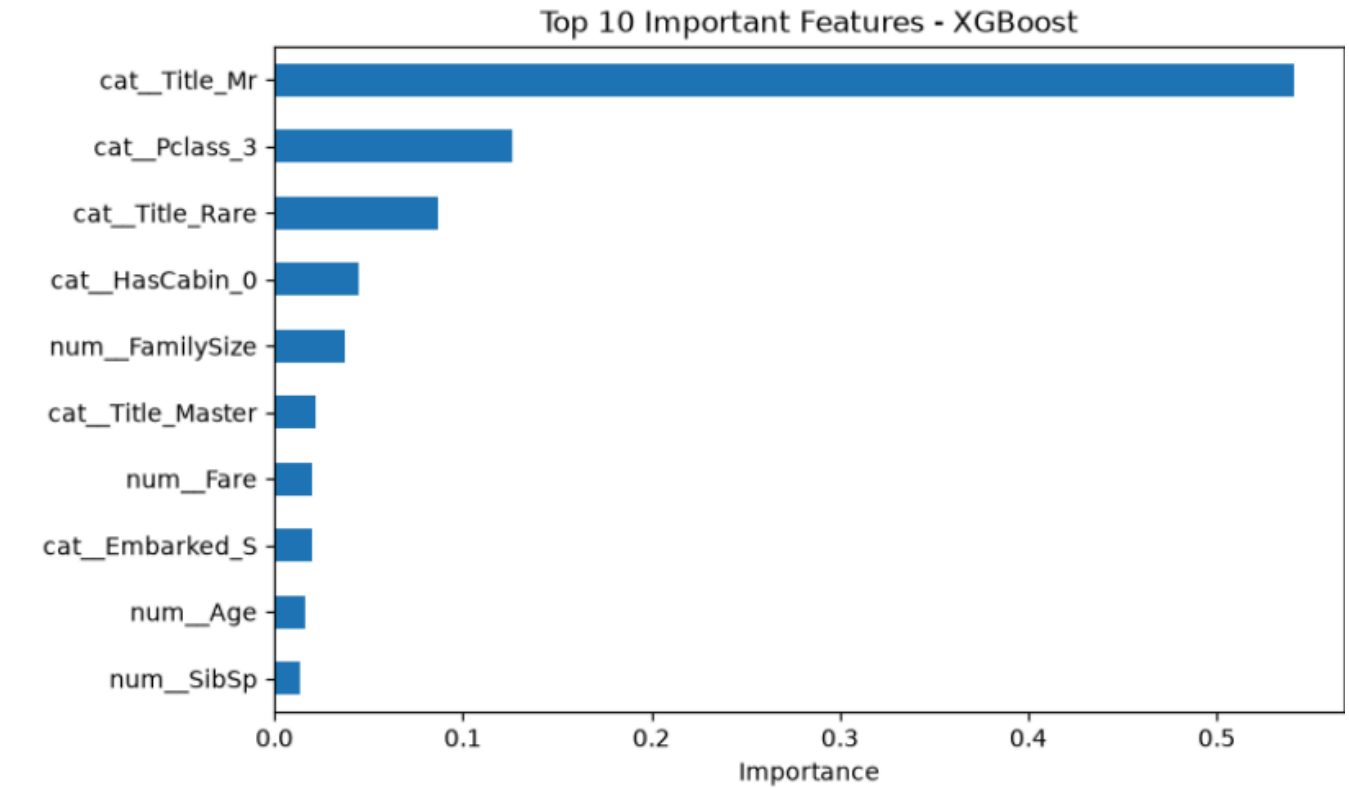
Confusion matrix



XGBoost is better at spotting deaths than survivors: it got 495 of 549 deaths right (90%) but only 257 of 342 survivors (75%). Its most common mistake was predicting "died" for 85 people who survived.

Feature importance

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plt.show()
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Title_Mr is by far the strongest feature (about 0.54): being an adult man was the clearest sign of not surviving. Third class and having no cabin come next, both linked to poorer passengers. Sex is missing from the top 10 only because Title_Mr already carries the same information, so the model relied on Title instead.

Conclusion and next steps

XGBoost is the best of the three models at 84.4% accuracy, because it learns combinations such as "3rd class and adult male" that a single straight line cannot.

Next steps:

- Tune XGBoost settings with GridSearchCV.
- Add features such as ticket group size or deck letter.

- Try to catch more survivors, the model’s main weakness.